

7.6 Classwork: Natural Logarithms and exp (no calculator)

1. Warmup

(a) Evaluate each of the following.

(i) $\log_9 27$

(ii) $\log_{27} 9$

(iii) $\log_{1/3} 27$

(iv) $\log_2 \sqrt[4]{8}$

(b) Let $a > 1$ and $b > 0$. Write each expression in terms of $\log_a b$.

(i) $\log_a(a^3 b)$

(ii) $\log_a\left(\frac{b^2}{a}\right)$

(iii) $\log_a \sqrt{a^3 b}$

(iv) $\log_{a^3} b$

2. Simplify each of the following. Give exact answers.

(a) $\ln(e^4)$

(b) $e^{\ln 7}$

(c) $\ln(e^{2x+1})$

(d) $e^{3\ln 2}$

3. Solve each equation for x . Give exact answers in terms of e or $\ln$.

(a) $e^x = 12$

(b) $\ln x = 5$

(c) $e^{2x-1} = 9$

(d) $\ln(3x + 1) = 2$

A graphing calculator is allowed for the remaining problems.

4. A scientist places 250 bacteria in a Petri dish. The number of bacteria N , t hours after the start of the experiment, is modelled by

$$N(t) = 250 e^{kt}, \quad k > 0.$$

After 3 hours, the dish contains 1000 bacteria.

- (a) Write down the value of N when $t = 0$.
- (b) Show that $k = \frac{1}{3} \ln 4$.
- (c) Find the number of bacteria after 5 hours. Give your answer to the nearest integer.
- (d) Find the time, in hours, at which the number of bacteria first reaches 10 000. Give your answer correct to three significant figures.
- (e) Find the doubling time of the population. Give your answer correct to three significant figures.

5. Lucia invests \$5000 in an account that earns interest at a nominal annual rate of 6 %, compounded continuously. The value of the account, in dollars, after t years is given by

$$A(t) = 5000 e^{0.06t}.$$

- (a) Find the value of the account after 4 years. Give your answer correct to the nearest cent.

- (b) Find the time required for the value of the account to first reach \$8000. Give your answer correct to three significant figures.

- (c) Find the time required for the value of the account to double. Give your answer correct to three significant figures.

- (d) A second account also begins with \$5000 but earns 6 % compounded annually. Find the value of this second account after 4 years, and state the difference between the two accounts at that time. Give both answers correct to the nearest cent.